AI-Based Fraud Detection System

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Abstract

This project develops a machine-learning system that scores banking transactions for fraud risk and flags suspicious ones for review. It combines imbalance-aware training, cost-based threshold selection and explainable predictions. All results come from synthetic data, not real bank data.

Problem Statement

Fraud is rare and changes over time, so fixed rules miss new patterns or raise many false alerts. Severe class imbalance biases models toward the majority class and makes accuracy misleading, while missed frauds and false alerts have different costs. Decisions must also be explainable.

Objectives

- Compare supervised and anomaly-detection models on highly imbalanced transaction data.
- Choose a decision threshold that balances missed frauds against false alerts.
- Provide real-time scoring with explainable predictions through an API and a dashboard.
- Evaluate recall, false positive rate and latency (targets: > 80%, < 5%, < 100 ms).

Technologies Used

Python, pandas, NumPy, SciPy, scikit-learn (Logistic Regression, Random Forest, Isolation Forest, MLP autoencoder), imbalanced-learn (SMOTE), XGBoost, SHAP, FastAPI, Streamlit, Docker, Prometheus, Grafana and boto3. XGBoost, SMOTE, SHAP, FastAPI, Streamlit and Docker are implemented in the source code but were not executed in the verification environment.

Methodology

- **Data:** synthetic dataset of 120,000 transactions (amount, timestamp, type, merchant category, country, device and IP risk scores, fraud label); chronological 70/15/15 train/validation/test split.
- **Preprocessing:** hour and weekday features, log amount, scaling and one-hot encoding.
- **Imbalance:** class weights, SMOTE (one Logistic Regression variant), XGBoost class weighting with randomized search.
- **Evaluation:** threshold minimizing an assumed cost (missed fraud 100, false alert 5) with validation FPR at most 4.5%; model chosen on validation data, reported on test data.
- **Prediction:** per-transaction scoring with optional SHAP explanation and PSI drift checks.

Key Features

- FastAPI endpoints for prediction (score, decision, risk level, SHAP explanation), drift and metrics.
- Streamlit dashboard: single-transaction scoring, batch analysis, performance reports and drift view.
- Drift detection (PSI, KS test), Docker packaging and Grafana alerts (not executed in testing).

Results

Measured on the synthetic test set (18,000 transactions, 147 frauds), run without XGBoost, SMOTE and SHAP (not installed). The selected Logistic Regression model achieved recall 49.7%, false positive rate 2.44%, precision 14.3% and ROC-AUC 0.856; Random Forest reached recall 41.5% at FPR 1.79%; Isolation Forest and the autoencoder were weaker (recall 12.9% and 1.4%). Scoring latency was 12.5 ms (99th percentile). The FPR and latency targets were met; the recall target was not. XGBoost, SHAP, API and dashboard results were not verified.

Conclusion

The project provides an end-to-end, explainable fraud-scoring pipeline. On synthetic data the baseline models did not reach the recall target, so tuned XGBoost results and validation on real transaction data are needed before deployment. Potential applications include bank transaction monitoring, payment gateways and e-commerce fraud review.